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Sustainability of Sal (*Shorea robusta*) forest in Bangladesh: past, present and future actions

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SUMMARY

Sal (*Shorea robusta*) forest is a threatened ecosystem in Bangladesh. Until the beginning of the 20th century Sal forests existed as a large continuous belt with rich biological resources, but increasing pressure has been placed on them since then due to the ever-increasing population. Most of the forest area at present is under occupation by encroachers and the remaining stands are stocked poorly. Biodiversity has declined rapidly and many animal species have become locally extinct. The Forest Department has established agroforestry and woodlot plantations as sustainable production system in the encroached and degraded forest area using a participatory approach. Some protected areas have also been established for conservation. Nevertheless, it is predicted that the present trend of management is inadequate and an intensive management policy is essential to restore the forest ecosystem. This paper explores trends of forest degradation, and past and present management initiatives, and recommends future management priorities.

Keywords: encroachment, management, change factors, agroforestry, sustainability

Durabilité de la forêt de Sal (*Shorea robusta*) dégradée du Bangladesh: actions passées, présentes et futures

M. ALAM,Y. FURUKAWA, S.K. SARKER et R. AHMED

La forêt de Shorea robusta est un écosystème menacé au Bangladesh. Jusqu'au début du XXème siècle, ces forêts existaient en ceinture continue, riche en ressources biologiques. Mais, la population en croissance constante plaça une pression extrême sur cet écosystème et changea le scénario. Le gros de la zone forestière est actuellement occupé par des intrus, et les peuplements restants sont de piètre qualité. La biodiversité a décliné rapidement et de nombreuses espèces se sont éteintes. Le département forestier a établi l'agroforesterie et les plantations de bois comme système de production durable dans la zone forestière envahie et dégradée, dans une approche avec participation. Certaines zones protégées ont également été établies pour la conservation. Les secteurs concernés anticipent néanmoins que le courant actuel de gestion est inadéquat et qu'une politique de gestion intensive est essentielle pour restaurer l'écosystème forestier. Cet article explore les courants de dégradation de la forêt, les initiatives de gestion présentes et passées, en incluant leurs faiblesses, et recommande pour finir les priorités de la gestion future.

Sostenibilidad del bosque de sal (*Shorea robusta*) en estado de degradación en Bangladesh: acciones pasadas, presentes y futuras

M. ALAM, Y. FURUKAWA, S.K. SARKER y R. AHMED

El bosque de sal (Shorea robusta) es un ecosistema amenazado en Bangladesh. Hasta principios del siglo 20, estos bosques formaban una zona continua rica en recursos biológicos, pero el aumento constante de la población ha ejercido una gran presión sobre este ecosistema y cambió el panorama. La mayor parte del área forestal se encuentra actualmente ocupada por parte de campesinos no autorizados y los árboles de las zonas que quedan son de baja calidad biológica. La biodiversidad ha entrado en fuerte declive y muchas especies se han extinguido. Basándose en un enfoque participativo, el Departamento Forestal ha establecido plantaciones agroforestales y lotes asignados de área fija para servir de modelo de sistema productivo sostenible en zonas forestales invadidas y degradadas. También se han establecido algunas zonas protegidas para su conservación. Sin embargo, algunos consideran que las pautas actuales de gestión son inadecuadas, y que hace falta una política de gestión intensiva para recuperar el ecosistema forestal. Este estudio analiza la tendencia a la degradación forestal y las iniciativas de gestión pasadas y actuales, incluyendo los problemas implícitos, y termina con algunas recomendaciones en cuanto a las prioridades para la gestión en el futuro.

INTRODUCTION

Bangladesh has an area of 147 570 km^2 and a population of 140 million (UNFPA 2007). Agriculture is the main occupation employing 68.5% of the labor force (BBS 2001). In terms of land use, 9.12 million ha are under cultivation, 1.91 million ha under public forest, 0.27 million ha under village groves, and 1.64 million ha remain constantly under water (Islam 2003). The remaining land (approximately 1.81 million ha) are occupied by tea gardens, fallow lands, rural and urban houses and ponds (Kibria *et al.* 2000). The public forest land, unclassed state forests and homegardens together make up about 17 percent (2.46 million ha) of the potential tree-growing area of the country (Ahmed 1999).

On the basis of geographical location, climate, topography, and management principles, the forests of Bangladesh can broadly be classified into Hill forests, Unclassed State Forests (USF), Plain land deciduous Sal (*Shorea robusta*) forests, Mangrove forests, Coastal forests and Homegardens (Khan 2003, GOB 2007, FMP 1993, FAO 1998, Rahman 2005). Bangladesh was rich in forest resources up to a few decades ago, but a rapidly expanding population, increased energy and wood consumption and overexploitation of resources has led to rapid degradation of forest reserves. A particular example of such degradation is the Sal forest which is surrounded by a high-density of population from all sides. This in turn has stimulated exploitation of the forest at an alarming rate and brought it close to extinction (Safa 2004).

DESCRIPTION OF SAL FOREST (*Shorea robusta*)

Extent, distribution, and biophysical settings

Sal forests are distributed mainly in South and Southeast Asia, occurring along the base of the tropical Himalayas from Assam to Punjab, in the eastern districts of Central India, and on the Western Bengal Hills (Figure 1). Sal forests have the widest distribution amongst all Dipterocarps, extending over an estimated area of 13 million hectares in India alone, with Bangladesh and Nepal together adding another one million hectares.

Broadly, Sal's natural range lies between the longitudes of 75° and 95° E and the latitudes of 20° to 32° N (Krishna and Nora 2006). Present biotic and abiotic features Sal forests are the results of actions and interactions of environmental and biotic factors over a long period and can be explained by theories of succession. A simple sketch of successional stages is shown in Figure 3 based on previous works of Jacob (1941), Champion and Osmaston (1962), Troup (1986) and Maithani *et al.* (1989) cited in Krishna and Nora (2006).

Sal forests constitute about 10% of the total forest land of Bangladesh. Until the beginning of the 20th century, these forests existed as a continuous belt from Comilla to Darjeeling in India. The present notified area of this forest is largely honeycombed with rice fields. FAO (1995)

FIGURE 1 *Natural zone of Sal forests (shaded area for Sal forests) after Stainton 1972, FAO 1985 and Krishna and Nora 2006*

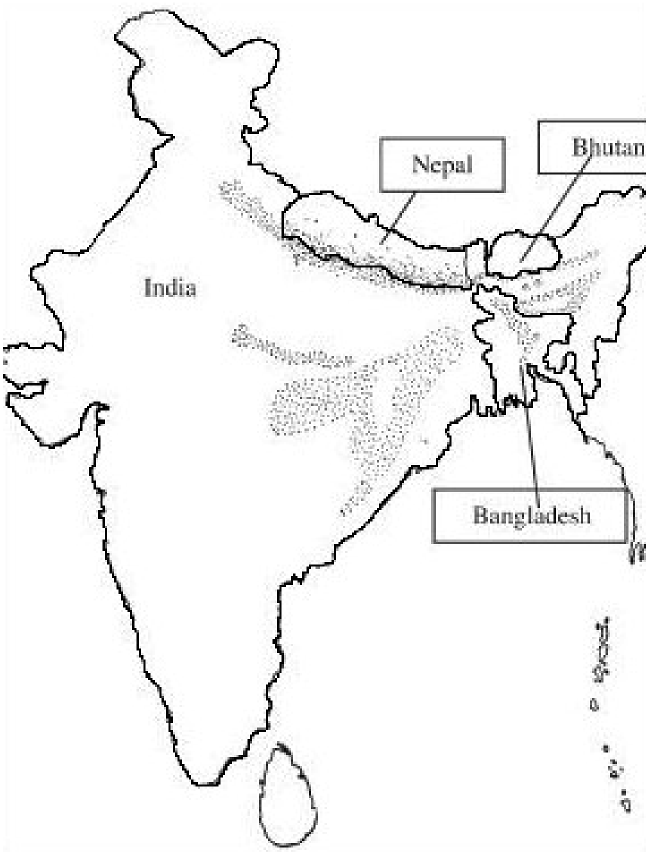


FIGURE 2 *Successional phases of Sal forests*

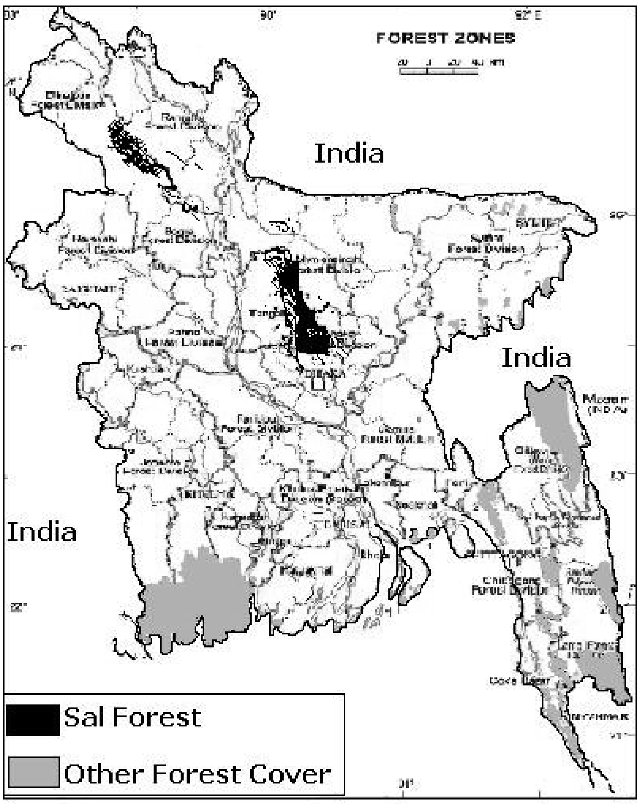
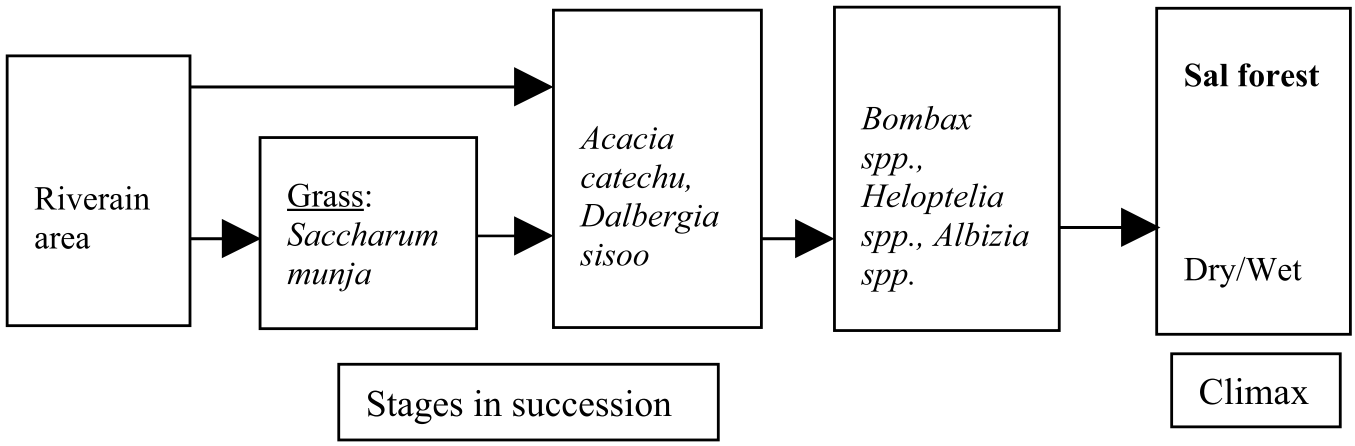


FIGURE 3 *Forest cover of Bangladesh including Sal forest*



estimated that about 36 percent of the forest cover existed in 1985, but more recent estimates suggest that only about 10 percent of the forest cover remains. A total area of 0.12 million ha is distributed over the central and north-western region of the country. About 86% of the total forest land is situated in the districts of Dhaka, Mymensingh, Tangail and Comilla (central region) with the remaining 14% in the greater districts of Rangpur, Dinajpur and Rajshahi (north-western region) (Figure: 1.2). The north-western region and Comilla district have little denuded scattered areas of forests at present. Of the total 122 012 ha forest land, 68 140 ha is reserved, 31 198ha is acquired, 2 689 ha is protected and 19 985 ha is vested (FMP 1992).

Deciduous forest is distributed mostly over the Madhupur Tract which lies between one and ten meters above the adjacent floodplains. The higher level lands are known as *chala* and the valleys, *Baid*. The climate of the Tract varies slightly from north to south, the northern reaches being much cooler in winter. Average temperatures vary from 28°C to 32°C in summer, falling to 20°C in winter. Rainfall ranges between 1000 mm and 1500 mm annually. Deciduous Sal forest in Bangladesh is also distributed over drought prone Barind Tract[1].

Sal forest resources

Sal forests are classified as tropical moist deciduous forests (Champion *et al.* 1965). FAO (2000) categorizes Sal forest into two subtypes, pure Sal and mixed Sal, on the basis of soil type and tree canopy. In the past pure Sal stands had a canopy that was nearly 100 percent and the growth of the trees was so rapid that these forests were considered inexhaustible (Khan 1998). Sal grew on shallow, dry, and less productive soils but such pure Sal forests now exist only in coppice form with sparse understorey and a relatively small number of species. Mixed Sal forests are dominated by Sal in the top storey but include many other associated species such as *Terminalia belerica*, *Dillenia pentagyna, Albizzia procera* and, *Lagerstroemia parviflora*. They grow on the deeper, moister and more productive soils of the Madhupur and Barind tract. The understory is more complex and includes a variety of deciduous and evergreen species (Table 1). The flora of this Sal forest type includes about 271 species of which 41 are tree species. Sal forests also include a high number of climbers and woody perennials of medicinal value.

Two centuries ago the forest was extremely rich in faunal diversity with elephant and rhinoceros having been reported, but they became extinct in the late nineteenth century. Leopards, bears (*Ursus thibetanus*), barking deer (*Muntiacus reevesi*), and many other animals which were abundant in the Sal forest areas have now disappeared although the leopard cat (*Prionailurus bengalensis*), fishing cat (*Prionailurus bengalensis*), jungle cat (*Felis chaus*) and small Indian civet (*Viverricula indica*) can still be found (UNEP 2001).

POLICY AND POLITICS OF LANDUSE: CHRONOLOGY OF PAST EVENTS

Prior to the 1940s, Sal forests were considered the private properties of *Zamindars* (local landlords) and other influential families (Chowdhury 1957). They were not particularly interested in the forests, except as an alternative source of revenue (Ahmad 1938), and attempted to maximize their revenue at the cost of over exploitation of the forests (Salam and Noguchi 2005). These forest areas were inhabited by a considerable number of tribal populations consisting of *Garos, Kooch, Hajongs and Mandhais* tribes (Ahmad 1938, Chowdhury 1957, Khan 1998). The landlords allowed the tribal people to do shifting cultivation within the forest tract. However, some forests in the central region had been under partial government management. As Sal (*Shorea robusta*) possessed excellent coppicing capacity,

[1] Barind Tract is distributed over north-western part of Bangladesh covering some districts of Rajshahi Division

TABLE 1 *Species composition in different strata of Sal forest*

Strata	Abundance (H=high; L=low)	Species
Top	H	*Shorea robusta, Albizia spp, Grewia laevigata, Lagerstroemia parviflora, Phyllanthus embelica, Terminalia belerica, Streblus asper*
	L	*Artocarpus lacucha, Cassia siamea*
Lower	H	*Leea crispa, Glycosmis arborea , Thespesia lampas, Careya arborea, Acacia pinnata*
	L	*Mimosa rubicaulis, Caessalpinia nuga, Miliusa veluttinal*
Grass	H	*Impreta cylindrica, Cynodon dactylon, Digitaria sanguinalis*
	L	*Chrysopogon aciculatus, Saccharum spontaneum*
Climber	H	*Mucuna pruriens, Ficus scandens, Smilax macrophylla, Asparagus acerosus*
	L	*Ichocarpus frutescens, Cryptolepis buchananii, Dioscorea bulbifera*

the system of management was a 'simple coppice system', supplemented by artificial regeneration where required (FMP 1992). The 'coppice with standard' system was also followed where mature trees were felled and the areas were protected for coppice regeneration (Hossain 1999). After the India-Pakistan partition in 1947, forest departments divided these forests into two working circles. One was a timber and conversion working circle where clear felling followed artificial plantation keeping the rotation to 70-80 years, and the second was a coppice working circle, with a rotation to 25 years. In 1987, the Forest Department began a project of rubber plantations in Sal forests under an Asian Development Bank (ADB)-funded project, and in 1988 about 7000 acres of Madhupur Sal forest was brought under rubber plantation. ADB had actively promoted the destruction of the Sal forests by financing projects for tree monoculture plantations using *Eucalyptus* and rubber among other species (Kabir and Ahmed 2005). Local people became concerned about gaining access to the goods and services they were used to obtaining from the Sal forest. Again, the government decided to establish a National Park in 8,436 ha area of Madhupur Sal forest. The indigenous *Garos* stood strong against the construction of boundarywall around the National Park because they took it as a barrier to their natural movement within the forest the construction of the walls was eventually suspended in the face of strong criticism and resistance.

Present landuses in this deciduous tropical forest include agroforestry, woodlot, and Sal coppice management, and recreation and conservation area management through establishment of ecoparks and national parks in different locations. These new management tools are bringing barren, degraded and encroached lands under forest cover by engaging local people in management. The chronology of major events in forest management in this deciduous forestland is as follows:

Before 1917: Management of forest by private owners.

1917: Appearance of first forest management plan for Bhawal Sal forest by J.R.P. Gent, the then D.C.F. of Bengal.

1934: Appearance of first management plan for Atia forest.

1950: Enactment of State Acquisition and Tenancy Act, 1950. Through this act the system of private ownership of forest became abolished and the government started taking control over the forestlands.

1956: The administrative jurisdiction of Dhaka Mymensingh Forest Division was separated and Mymensingh Forest Division was created. Which again was divided into Tangail and Mymensingh Forest Division? Fresh and separate working plans were prepared for the forests under the jurisdiction of these administrative units.

1959: Forest Department prepared a management plan in this year for the forest areas of Dinajpur, Rangpur and Rajshahi. The plan prescribed three working circles: conversion working circle, coppice working circle and afforestation. working circle.

1962: Establishment of Madhupur National Park (area 8436 ha) for conservation,research, education and recreation.

1969: Tangail forest Division was established with the forestland and forest area of Ghatyle, Kalihiti, Mirjapur and Shokhipur of Tangail district from Mymenshing.

1972: Enforcement of moratorium on all kind of extraction from the forest. Consequently, very little management practices have been carried out in the forested areas since then.

1974: Bhawal National Park in Gazipur was established and maintained as a national park, but not declared officially until 1982 under the Bangladesh Wildlife Act, 1974

1976: The plan was revised, though did not work, to create two working circles: community forestry working circle and commercial working circle.

1982: Establishment of Bhawal National Park (area 5 022 ha) for conservation, research, education and recreation.

1989: Initiation of participatory forestry (social forestry) programmeme by the forest department in the name of agroforestry and woodlot plantation under Thana Afforestation and Nursery Development Project (TANDP).

1992: Preparation of Forestry Master Plan that properly addressed future management priorities in Sal forest.

1994: Promulgation of latest forest policy with a provision-

"State-owned hill and Sal forests will be managed as production forest except those declared as ' Protected Areas ' for preserving soil, water and biodiversity. The production forests will be managed on a commercial basis with due consideration to the environment"

1997: Forestry Sector Project for 1997–2004 was initiated. The programme proposed to establish woodlot and agroforestry plantations on degraded Sal forests

2001: Harvesting of mature trees in the participatory forestry plantation began in Sal forest

CHANGE FACTORS AND THE NATURE OF DEGRADATION

During the times of private ownership of Sal forest the general policy of the proprietors was to collect as much revenue out of the forest as they could, with no concern for sustainability. However, in some areas parts of the forest were leased out for a specified period of 5 years and the lessee was permitted to remove of all trees above 1 ft. 6 inches in girth above 3 ft. from the ground. This practice did great damage to the quality of the forest. All good quality trees were being cut leaving only poor ones and no attention was given to ensure future regeneration (Alam 2006)). This resulted in a very poor stocking of forest. Inappropriate administrative control by landlords, coupled with the practice of shifting cultivation resulted in numerable patches of vacant land devoid of any tree growth. The government documents consistently condemned the practice of shifting cultivation and identified it as "the greatest enemy of forest" (Khan 1998).

At present the majority of forest dwellers are *Garo* who migrated from India at least one hundred years ago. In addition other tribes such as *Koch* also came to settle during the same time (Pant 1991 quoted in Ahmed *et al.* 2007). The Bengali-speaking people, who used to live along the fringes of the extensive forests, entered gradually into the forests as residents. These populations led to drastic ecosystem changes in the last thirty years whereby more than 60% of the Sal forest was relatively densely wooded 30 years ago but today, the forest has been reduced both in extent and tree density as well as stand quality (Chowdury 1994, Safa 2004).

Bangladesh is a land of huge population pressure with an average population density of about 750/km^2, reaching some 1300/km^2 near Dhaka, Chittagong and other population centers (FAO 2000), this high density of population creates 'diffusion pressure' towards the forests resulting in settlement near or inside the forests. Expansion of cities towards forest with unplanned urbanization policy as well as high density of population just outside the forest is creating tremendous pressure resulting in illicit removal of forest resources as well as encroachment of forestland. Legal gaps due to past tenurial history of the lands, intricate nature of the boundary of the forestland and cultivable lands and failure to complete the forest settlement operations initiated in the 1950's have been enhancing the chance of encroachment of forest property. Existence of organized groups of people who indulge in illicit cutting and removal of valuable trees of the forests are leaving the Sal forest in a denuded state. Sal forests in the Barind Tracts also bear the marks of extreme human interference where people have cut back the Sal coppice to such an extent that the stumps have lost coppicing power (FMP 1992).

PRESENT MANAGEMENT: VARIOUS PRODUCTION PRACTICES

Agroforestry

The Forest Department, with financial assistance from the Asian Development Bank, initiated woodlot plantation and agroforestry under the Thana Afforeststion and Nursery Development Project (TANDP) in November 1989. The project proposed to establish 238 ha of agroforestry plantations in Sal forest, which has subsequently become an important element of forest management. The Forest Department (FD) allocated plots (usually 1.0 ha) among each of the selected participants. After getting a plot from the FD the participant first made a rough boundary around the allocated plot then cleared the ground prior to cultivation. After initial fertilization, some participants planted paddy as the first agricrop, while others sowed mustard seed. After producing the first crop, zinger, papya, palmkin and, pineapple were cultivated. At the same time, tree crops were planted in the plot. Usually agricultural crops were practiced up to the fourth year of the rotation until the crops could no longer gain adequate light under the trees. The FD selected the species of trees to be planted in the agroforestry plots and supplied seedlings of Akashmoni (*Acacia auriculiformes*), Bokain (*Melia azadarach*), Mangium (*Acacia mangium*) and, Gamar (*Gmelina arborea*) for plantation in the plots. All of the first thinning benefit goes to the participant but the 2nd thinning and final harvest are divided into a predetermined ratio (Table 2).

TABLE 2 *Benefit sharing of shareholders of agroforestry cultivation*

Benefit from	Shareholder	Share (%)
1st thinning	Participant	100
2nd thinning, intermediate returns and final harvest	Participants	45
	Forest Department	45
	Tree Farming Fund (TFF)	10

TABLE 3 *Benefit sharing of shareholders of woodlot plantation*

Benefit from	Shareholder	Share (%)
1st thinning	Participant	100
2nd thinning, intermediate returns and final harvest	Participants	40
	Forest Department	50
	Tree Farming Fund (TFF)	10

Woodlot

The FD initiated woodlot plantations under the Thana Afforeststion and Nursery Development Project (TANDP) in November 1989 with the objective to establish 16 188 ha of woodlot plantation in the Sal forest. Salam and Noguchi (2005) calculated that during 2002-2003 the mean production was 513 poles, 767cft small sawlogs and 1 249 cft fuelwood. Sharing of the final harvest followed the ratio shown in Table 2. The principle objective of the woodlot was to grow short and medium rotation trees that will meet the local demand of fuel and timber, but Alam (2005) revealed that the participants show greater preference for agroforestry than woodlots.

Sal coppice management

The silvicultural system applied in Sal forests was a coppice with standard regime. In this system mature trees were felled and the areas protected for coppice regeneration on a rotation of 25 years (Hossain 1999). Areas where Sal (*Shorea robusta*) trees are comparatively fewer are managed under a clear felling system followed by artificial regeneration with Sal (*Shorea robusta*) or other suitable species. The main feature in almost all of the working plans was to divide the forest into several working circles, e.g. timber working circle, conversion working circle and coppice working circle depending on the crop condition. The prescription was to follow clear felling with artificial regeneration (mainly with Sal and other suitable species) in the timber and conversion working circles with a rotation of 80 years.

The existing Sal stands are in a critical situation and with declining growing stock due to over exploitation, low genetic base and traditional management practices. As a means of addressing this situation, the concept of participatory management has been introduced in Sal coppice management whereby an isolated patch or block of Sal forests is assigned to a user group of 20-35 members formed from the nearby village. The members of the user group are target beneficiaries who area mainly landless people, poor householders and destitute women. Each participant is provided with training on Sal coppice management, including thinning, pruning and weeding.

Recreation and conservation area management

The Government declared 8 436 ha area of Sal forest situated in Madhupur as a National Park in 1962 with the aim of conserving forest genetic resources as well as establishing forest based recreation facilities. At present it houses a total of 176 species of plants and was once very rich in wild fauna such as elephant, buffalo, tiger, leopard and, peafowl but these are extinct now along with another 21 species of mammals, 140 species of birds and 29 reptiles (GOB 2007). In 1974 the Government declared a second National Park in the Sal forest, the Bhawal National Park, which is 5 022 ha in area and just 40km from the capital city, Dhaka.

IMPACT OF PRESENT MANAGEMENT

Impact on local livelihood

Agroforestry and woodlot production have become income-generating activities for poor forest dwellers of Sal forest area. A study by Ahmed *et al.* (2007) revealed that agroforestry in degraded Sal forests is a highly productive and economically beneficial land use system. During field survey the authors estimated that total cost incurred by a farmer for cultivation of 1 ha plot is US$ 1 070. The total revenue after 4th year from selling agricultural crops becomes US$ 178; trees provide a further revenue of US$ 11 030 (625 trees @ US$ 16.64 each). Gross revenue from thinning and pruning was estimated to be US$ 6 213. Muhammed *et al.* (2007) estimated in their study that the Net Present Value (NPV) and Benefit-Cost Ratio (BCR) of the agroforestry plantation were US$ 28,255 and 1.64 respectively. Moreover, Ahmed, *et al*. (2007) concludes that under the circumstances of excess population pressure, poverty and scarcity of land resources in Bangladesh agroforestry can be the best solution to earn money and ensure a scope of income generation for forest dependent people. Safa (2004) attempted a study to examine the change in socio-economic structure due to participation in agroforestry and woodlot programmes. The study indicates that participatory management and agroforestry have had a positive impact on settlers' livelihoods as well as on sustainability of forest resources.

Impact on forest sustainability

Effective sustainable forest management requires that the various functions of forests be considered within each stand and across various stands at the landscape level (Fujimori 2001). The ultimate objective of sustainable forest management is to meet the forest-related goals and needs of the current generation without compromising the ability of the future generations to meet their own needs. The participatory approaches of agroforestry, coppice management and woodlot management are regarded by many authors as important tools of sustainable forest development and management in degraded Sal forest (Safa 2004, Salam and Noguchi 2005, Alam 2006, Chowdhury 1994, Muhammed *et al.* 2007). Another approach to management in Sal forests is the establishment of ecotourism areas, which have already proved to be an effective tool for nature conservation in other parts of the world. Ecotourism has three important criteria: conservation of nature and natural resources, meaningful community participation, and it is profitable and sustainable.

Impact on change in forest cover

The objectives of participatory Sal coppice and conservation area management are not to increase the forest cover, rather these management tools are aimed at reducing the vulnerability of forestland from encroachment and denudation. However, management in terms of agroforestry and woodlots is being practiced by the forest department on degraded, denuded and encroached forestlands with consequent increases in forest cover in those areas. The FD established plantations on denuded and degraded forest land under two projects (TANDP in 1989-1994 period and FSP in 1997-2004 period) producing a total area of plantations (agroforestry15, 143 ha and woodlot 36, 974 ha) developed under these two projects of 52, 117 ha (0.052M ha). Since the Forestry Sector Project for 1997–2004 established plantations on denuded forests as well as on areas where trees from a previous programme (first rotation) had already been harvested (Salam and Noguchi 2006), the net estimated increase in forest cover is about 0.04M ha, which is one third of total land area (0.12M ha) of Sal forest.

PRESENT MANAGEMENT DRAWBACKS

The emphasis of previous management plans and policies of Sal forest was on increasing productivity by creating high value of timber stock through clear felling the existing natural stands followed by artificial regeneration with exotics or fast growing species. Financial return was the main objective and even now some of the management strategies reflect the same objectives as in the past. There is, at times, little concern for the livelihoods of forest dependent local and ethnic communities.

The woodlot plantation management scheme with various exotic species (*Eucalyptus spp., Acacia spp*.) began at the same time in the Sal forest under the Thana Afforestation and Nursery Development Project (TANDP) with the objective of to meet the fuelwood requirement of the local communities. The main criticism of woodlot plantations was that they had threatened the habitat of indigenous forest communities living in the forest. But in addition, severe ecological problems were caused by woodlots because, in many places throughout the forest, coppices of Sal trees and other indigenous species were clear-cut for the preparation of woodlot blocks. This destroyed the possibility of regeneration of natural forest in many areas. Moreover, the impact of clearing the forest for woodlot and rubber plantation affected particularly faunal diversity of that area, as these trees do not support wildlife because they do not produce edible fruit or nectar for them (Ameen 1999).

The antagonistic relationship between the FD and local people is a big obstacle confronting Sal forest management (Gain 1998), with conflict existing for some time between the FD and ethnic communities.

Banana cultivation is widely practiced in different locations of the forest due to the high and quick economic returns available to forest dependent people. But indiscriminate use of fertilizers, pesticides and other chemicals are decreasing drastically the natural fertility and productivity of soil, and soil erosion due to the banana production system has become a serious problem.

PRIORITIES FOR THE FUTURE

The initiatives undertaken presently by the Forest Department have added new dimensions to Sal forest management and there is a good opportunity to restore the ecology, diversity, and productivity of the forest to its previous levels. The following tasks are presented as future priorities.

Production

- Sal forest is a natural home for *Shorea robusta*, therefore major attention should be placed on increasing its productivity rather replacing it with other species. A study quoted in Rahman *et al.* (2005) indicated that the protected Sal coppice can be much more productive than block plantation per hectare basis .In addition, studies from West Bengal indicated that forest protection costs are only 5% of the cost of plantation establishment (West Bengal Forest Department 1989).
- Agroforestry should be given priority over woodlots where possible. Different studies (e.g. Ahmed 2007, Safa 2004, Muhammed *et al.* 2007)) have indicated that agroforestry is financially more viable than woodlot management.
- The FD introduced bamboo in Sal forest recently, and Safa (2004) also suggested bamboo farming in this forest. But the bamboo introduction did not seem to have any positive impact at all, therefore, a detailed investigation is required before introducing bamboo.

Protection

- Encroachment and illicit removal of forest resources can be reduced to an extent by demarcation and protection of forest land through boundaries, brick pillar, canal excavation and buffer zone management. Moreover, appropriate measure must be taken to resolve tenure issues and conflict between public and private titles.
- The real importance for conservation of deciduous forest lies in the fact that they are the most used and threatened ecosystem (Janzen 1986). Protection zones in the name of National Parks and Eco Parks can be expanded and

more recreational facilities can be provided to tourists by maintaining the unique biodiversity of the forest.

- Where possible, replanting/vacancy filling with *Shorea robusta* and its natural associate species like *Terminalia bellerica*, *Dillenia pentagyna, Albizzia procera* and, *Lagerstroemia parviflora* should be examined. Naturally the Sal forest in Bangladesh harbours a large number of species of medicinal plants. Identification, protection and detailed study of the state of medicinal plants of the forest is also required.

Socio-economic

- The existing conflict between ethnic communities and the FD needs to be resolved. If the conflict persists, it will compromise all of the sustainable management efforts over the long term. Participation of ethnic people in participatory forestry activities is very low (Ahmed *et al.* 2007), and this needs to be addressed.
- NGOs have added a new dimension to forest management, which has ensured community participation and protection of the forests (Safa 2006). They can play an important role in motivation and awareness building among the villages regarding the biodiversity of forest and its sustainable utilization. Moreover, they can support people with micro-credit facility in developing forest based small-scale enterprises.
- The decision-making process in Bangladesh is a ‘‘top-down’’ approach where decisions are handed over to local stakeholders from the top authorities. The scenario should be changed and active participation of stakeholders in decision making should be required.

CONCLUSION

Under the circumstances of high-density population pressure, poverty and scarcity of land resources in Bangladesh, it is essential that any available land be intensively utilized through sustainable development while maintaining an ecological balance. Sal forest is the only plain land forest in Bangladesh and is important from both national economic and environmental perspectives. Only wise, scientific, and intensive management can save the forest land and its resources. Future managers should aim at managing Sal forests simultaneously for multiple products and ecological sustainability as well as for the socio-economic benefit of poor forest dependent people living in and around the forest area. Managing forests with such aims will earn money, save the land from encroachment, ensure a scope of income generation, and be a model of sustainable land management.

REFERENCES

AHMAD, Y.S. 1938. *Working plan for the Dacca-Mymensingh forest division (1937-38 to 1957-58).* Bengal Government Press, Alipore.

AHMED, F.U. 1999. Summary of the social forestry programmes of the Bangladesh Forest Department. *The Bangladesh Agroforestry Newsletter* 9:1-2.

AHAMED, F. U. 1993. *Respondents to environmental degradation: Some implications of a social forestry project in Bangladesh.* M. Phil. dissertation, Cambridge University, Cambridge.

AHMED, R., UDDIN, M.B. and ALAM, M. 2007. Participatory Agroforestry Production Model in Degraded Deciduous Forest: A Sustainable Landuse Experience from Bangladesh. *International Journal of Sustainable Development and World Ecology*. (communicating)

ALAM, M. 2006, *Management plan for Madhupur area under Tangail Forest Division for the period 2005-2015.* Department of Forestry, Shahjalal University of Science and Technology, Sylhet.

ALAM, R. 1998. P*articipatory approach in degraded Sal forest in Gazipur district.* Review paper submitted to Institute of Forestry and Environmental Sciences, University of Chittagong.

AMEEN, M. 1999. *Development of guiding principles for the prevention of impact of alien species.* Paper presented at a preparatory workshop for the 4th meeting of SBSTTA to CBD, organized by IUCN Bangladesh, Dhaka, 25 May, 1999.

BBS. 2001. *2001 Statistical Year Book of Bangladesh.* Bangladesh Bureau of Statistics. Government of People’s Republic of Bangladesh.

BANGLAPEDIA, 2007. *National Encyclopedia of Bangladesh*. Asiatic Society, Dhaka. http://www.banglapedia.org/ accessed on January 24, 2007.

CHAMPION, H and OSMASTON, F.C.1962 *E. P.Stebbing’s The Forests of India. Volume IV*. Oxford University Press, London.

CHAMPION, H.G., SETH, S.K. and KHATTAK, G.M. 1965. *Forest Types of Pakistan*, Pakistan Forestry Institute, Peshawar.

CHOWDHURY, M.R. 1957. *Working scheme for Mymensing division 1955-56 to 1959-60*. East Pakistan Government Press, East Pakistan.

CHOWDHURY, R. A. 1994. History and Importance of Sal Forests and Current Management Status, *In*: Agro forestry for the degraded ‘Sal’ forest, Proceedings of national workshop held at Bangladesh Agricultural Research Council, BARC and Forest Department with the financial assistance of APAN, Dhaka.

DOUGLAS, J.J. 1981. *Consumption and supply of wood and bamboo in Bangladesh*. UNDP/FAO/GOB project BGD/78/010. Field document no.2, Food and Agricultural Organization, Rome.

EUDEY, A. 2000. *Trachypithecus pileatus.* In: IUCN 2003. *2003 IUCN Red List of Threatened Species. www.redlist.org*. Downloaded on 28 August 2004.

FAO. 2000. *FRA 2000 Forest Resource of Bangladesh Country Report.* Food and Agriculture Organization, Rome.

FAO. 1993. *Forest resources assessment 1990: Tropical countries*. FAO Forestry Paper 112, Rome.

FAO. 1998. *Woodfuel in Bangladesh-production and marketing*. Food and Agricultural Organization of the United Nations, Bangkok.

FAO. 1995. *Forest resources assessment 1990: Global Synthesis.* FAO Forestry Paper 124, Rome.

FMP. 1992. *Forest Management*. Forestry Master Plan. Ministry of Environment and Forest, Government of Bangladesh. UNDP/FAO BGD 88/025.

FUJIMORI, T. 2001. Ecological and silvicultural strategies for sustainable forest management, Elsevier, Tokyo.

GAIN, P. 1998. *A Journey through a dying heritage: The last forest of Bangladesh.* Society for environment and Human Development. Dhaka.

GOB. 2007. Forest Department official website: http://www.bforest.gov.bd/agro.php, Dhaka: Government of People's Republic of Bangladesh. Accessed January 25, 2007.

HOSSAIN, S. 1999 Participatory forest management in Bangladesh. http://www.iges.or.jp/en/fc/phase1/2ws-3-Bangla.pdf.

ISLAM, S.S. 2003. Status of forest genetic resources conservation and management in Bangladesh. *In:* LUOMA-AHO, T., HONG, L.T., RAO, V.R. and SIM, H.C. (eds) Proceedings of the Asia Pacific Forest Genetic Resources Programmeme (APFORGEN) Inception Workshop, Kepong, Kuala Lumpur, Malaysia,15–18 July, 2003.

JACOB, M.C. 1941. Sal regeneration de novo a rejoinder. *Indian Forester.* **67** , 337 – 346.

JANZEN, D.H. 1986. *Biodiversity* (ed. Wilson, E.O.), National Academy Press, Washington, pp. 130-137.

KABIR, D.S. AND AHMED, A.Z. 2005. Wildlife biodiversity in Bhawal National Park: Management techniques and drawbacks of wildlife management and nature conservation. *Our Nature*, (2005) 3: 83-90.

KHAN, M.A.R. 1981. The non-human primates of Bangladesh. *Tiger Paper*. Vol. **8**(1), 12-15.

KHAN, N.A. 2003. *Regional study of forest policy and institutional reform final report of the Bangladesh case study*. Asian Development Bank, Mandaluyong City.

KHAN, N.A. 1998. *A political economy of forest resource use: Case studies of social forestry in Bangladesh*. Ashgate Publishing Ltd. Aldershot.

KIBRIA, M.G., SARKAR, D.C., HOSSAIN, M.A.T., MANNAN, M.A., MOTALEB, M.A. and ISLAM, S.S. *2000 Forest statistics of Bangladesh*. Forest Economics Division, BFRI.bulletin 4, Chittagong.

KRISHNA, H.G. and NORA, N.D. 2006. Ecological and anthropogenic niches of Sal (*Shorea robusta* Gaertn. f) forest and prospects for multiple-product forest management-a review. *Forestry*, 79(1): 81-101

MAITHANI, G.P., SHARMA, D.C. and BAHUGUNA, V.K. 1989. Problems of sal forests – an analysis. *Indian Forester.* 115 , 513 – 525.

MUHAMMED, N., KOIKE, M., HAQUE, F. and MIAH, M.D. 2007. Quantitative assessment of people-oriented forestry in Bangladesh: A case study in the Tangail forest division. *Journal of Environmental Management*, doi:10.1016/j.jenvman.2007.01.029

PANT, M.M. 1991. *Agroforestry in Bangladesh.* Chittagong: Institute of forestry and Environmental Sciences.

QUDDUS, A.H.G., ALI, S.M.I., BHUIYAN, A.M.A. and HOSSAIN, M. 1992. *Greening the hills: the Betagi-Pamora agoroforestry experience. Research report series 1*, BARC-Winrock International, Dhaka.

RAHMAN, M.M. 2005. Management and economics of homestead forests in Bangladesh. PhD thesis submitted to United Graduate School of Agricultural Sciences, Ehime University, Ehime.

RAHMAN, M.M., ALAM, M. and RAHMAN, M.M. 2005. Diversity of Forest Tree Species of Moist Deciduous Forest of Madhupur Tract of Central Bangladesh. (Unpublished manuscript)

SAFA, M.S. 2006. The role of NGOs in improving social forestry practice: Do they promote livelihood, sustainability and optimal land use in Bangladesh? *Small-scale Forestry*, 5(2): 207-229.

SAFA, M.S. 2004. The Effect of Participatory Forest Management on the Livelihood and Poverty of Settlers in a Rehabilitation Programme of Degraded Forest in Bangladesh. *Small-scale Forest Economics, Management and Policy*, **3**(2): 223-238.

SALAM, M. A., and NOGUCHI, T. 2005. On sustainable development of social forestry in Bangladesh: Experience from Sal (*Shorea robusta*) forests. *Environment, Development and Sustainability* (2005) **7**: 209–227.

TROUP, R.S. 1986. *The Silviculture of Indian Trees.* International Book Distributors, Dehradun, India.

UNEP 2001. *State of the environment report – Bangladesh.* United Nations Environment Programmeme, pp138.

UNFPA 2007. United Nations Population Fund, Bangladesh. http://www.unfpa-bangladesh.org. Accessed December 11, 2007.

WENGER, K.F. (ed). 1984. *Forestry handbook (second edition).* John Wiley and Sons, New York.

WEST BENGAL FOREST DEPARTMENT. 1989. *Forest Regeneration Through Community Protection.* Proceedings of the Working Group Meeting on Forest Protection Committees, Calcutta, West Bengal.